

Project Style Guide and Glossary

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1 Style Guide

1.1 Purpose

This document defines naming, formatting, and style conventions for reports, scripts, metadata, and documentation used in the project. Additionally, the document acts as a dictionary of project-specific vocabulary.

1.2 Terminology and Naming Conventions

1.2.1 Macrozone Names

Macrozone	Preferred Term	Notes
tallgrass	Tallgrass Prairie	No hyphen; conventional in literature
mixed	Mixed-Grass Prairie	Hyphenated compound modifier
shortgrass	Shortgrass-Steppe	Hyphenated; “Prairie” is redundant

Use hyphenated forms when modifying a noun (e.g., “Shortgrass-Steppe region”), and unhyphenated when used as a noun (e.g., “We sampled the Shortgrass Steppe”).

1.3 Variable Naming

1.3.1 General Use Cases

- Use snake_case for all variable names (e.g., `ppt_ann_mm`, `slope_pct`)
- Place `site_no`, `latitude`, and `longitude` as the first columns in exported datasets
- Use standard units in suffixes (`_mm`, `_C`, `_pct`, `_km2`)

1.3.2 Specific Use Cases for Spatial Data Workflows

The project strives to use explicitly named, project-specific ID fields, which are treated as a UID. One qc check on asserting UID status prior to a join is by using the `assert_inputs_ok()` helper function. The reason UIDs are important is to ensure one to one cardinality in joining or merging datasets.

- **site_no** — USGS gage ID (UID, not regenerated).
- **macro_id** — persistent UID for macrozones (sequential, unique).
- **catch_id** — NHDPlus catchment UID.

1.4 Writing Style

- Use full phrases before acronyms on first use (e.g., “Normalized Difference Vegetation Index (NDVI)”).
- Follow USGS and EPA naming conventions when describing datasets, flowlines, and catchments.
- Avoid unnecessary repetition, in other words tautologies. For example, avoid writing “Shortgrass-Steppe Prairie”.

1.5 Citation Style

- Internal datasets: reference relative path in `docs/metadata/`.
 - External datasets: provide DOI or citation in README/data dictionary.
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2 Glossary

2.1 Coordinate Reference System (CRS) Codes

CRS (Coordinate Reference System) refers to the general concept of a map coordinate system. A CRS includes: datums, projections, units, transformations, and other geodetic parameters.

- A datum is an earth model, ellipsoid or spheroid, and the origin point for the system
- The map projection describes how the curved Earth is flattened onto 2D.

EPSG codes are unique numeric identifiers for specific CRS, in other words, EPSG is a registry and naming system for CRS. EPSG codes are standardized, unique integer numbers (e.g., EPSG:4326) used to identify a specific CRS definition. The acronym comes from the EPSG Geodetic Parameter Dataset, which was originally compiled by the European Petroleum Survey Group (now defunct) and now maintained by the International Association of Oil & Gas Producers (IOGP). EPSG codes provide a consistent way to refer to and specify different CRS and transformations. Some EPSG codes used in the project include:

- EPSG:4269 – NAD83 / unprojected (GRS 1980 ellipsoid) in lat lon.
- EPSG:4326 – WGS84 / unprojected (WGS 1984 ellipsoid) in lat lon. EPSG:4326 is used by the National Elevation Dataset (NED).
- EPSG:5070 – NAD83 / Conus Albers (GRS 1980 ellipsoid) in meters. **EPSG:5070 is the project standard EPSG code.**

2.2 An Explanation of ID-term in Spatial Data Workflows

- **ID (Identifier)** is the *generic term* for a field that uniquely (or semi-uniquely) identifies records, and often consists of sequential integers assigned on export. While it is commonly used in shape-files/GeoPackages (ID, id, FID), it *may not be stable across exports or joins*.
- **UID (Unique Identifier)** is a term that explicitly means *guaranteed unique* within the dataset. A UID provides a stronger guarantee than plain ID, meaning no duplicates and no recycling. A UID is often in a UUID/GUID format (e.g., 550e8400-e29b-41d4-a716-446655440000) or carefully managed integer sequences. UID is *useful when performing merge/join across multiple datasets*, and when persistence across time and versions is needed

- **OID (Object ID)** is a term from ESRI geodatabases and is an automatically maintained field (OBJECTID or OID). OID is internal to the geodatabase and ensures every feature has a unique row identifier. *OID is **not** stable upon re-export or recompute*, as Esri may regenerate an OID. Therefore, it should not be used as an analysis join key in persistent workflows.